Technical Assessment

Complete Technical Assessment

Purpose: This assessment is designed to prepare students for real-time technical interviews in Data Analyst, SQL Developer, and Database Developer roles.

Assessment Instructions

- Total Questions: 5
- Difficulty: Intermediate to Advanced
- Each question carries 20 marks.
- Create the required tables and insert the given records.
- Use only the tables provided for each question.
- Write queries based on the business requirements.
- Avoid hardcoding results wherever possible.
- Use appropriate joins, subqueries, CTEs, aggregate functions, and window functions.

Important: Each question is an independent real-time scenario.

1. Database Setup – Common Tables

Use the following three tables for all five assessment questions. Each table contains five records.

Table 1: departments

Business purpose: Stores department information in an organization.

```
None
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);

INSERT INTO departments VALUES
(1, 'IT', 'Chennai'),
(2, 'HR', 'Bangalore'),
(3, 'Finance', 'Chennai'),
(4, 'Sales', 'Mumbai'),
```

```
(5, 'Marketing', 'Hyderabad');
```

Table 2: employees

Business purpose: Stores employee details, salaries, department information, and reporting managers.

```
None
CREATE TAbLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE,
    salary DECIMAL(10,2),
    dept_id INT,
    manager_id INT,
    joining_date DATE,
    city VARCHAR(50),
    bonus DECIMAL(10,2),

    FOREIGN KEY (dept_id)
    REFERENCES departments(dept_id),

    FOREIGN KEY (manager_id)
    REFERENCES employees(emp_id)
);

INSERT INTO employees VALUES
(101, 'Arun', 'arun@gmail.com', 75000, 1, NULL, '2021-01-15',
'Chennai', 10000),
(102, 'Priya', 'priya@gmail.com', 55000, 2, 101, '2022-03-20',
'Bangalore', NULL),
(103, 'Karthik', 'karthik@gmail.com', 65000, 1, 101,
'2020-06-10', 'Chennai', 8000),
(104, 'Divya', 'divya@gmail.com', 40000, 3, 103, '2023-07-05',
'Mumbai', 5000),
(105, 'Rahul', 'rahul@gmail.com', 35000, 4, 103, '2024-02-12',
'Hyderabad', NULL);
```

Table 3: projects

Business purpose: Stores project assignments and project budgets.

```
None
CREATE TABLE projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100),
    emp_id INT,
    project_budget DECIMAL(12,2),
    start_date DATE,
    status VARCHAR(30),

    FOREIGN KEY (emp_id)
    REFERENCES employees(emp_id)
);

INSERT INTO projects VALUES
(201, 'Banking Analytics', 101, 500000, '2024-01-10',
'Completed'),
(202, 'HR Automation', 102, 200000, '2024-03-15', 'Ongoing'),
(203, 'Customer Prediction', 103, 450000, '2024-05-20',
'Ongoing'),
(204, 'Financial Dashboard', 104, 150000, '2024-07-01',
'Completed'),
(205, 'Sales Forecasting', 105, 300000, '2024-09-10', 'Pending');
```

2. Technical Assessment

Question 1 – Employee Performance and Salary Analysis

20 Marks
HR Analytics Scenario

Scenario:

A company wants to analyse employee salary distribution and identify employees who are earning more or less than the average salary of their department.

The HR team also wants to generate a salary report for management.

Tasks:

1. Display employee name, department name, salary, and location using an INNER JOIN.

2. Find employees whose salary is greater than the average salary of their respective departments using a correlated subquery.
3. Display the total salary, average salary, minimum salary, and maximum salary for each department using GROUP BY.
4. Display only those departments whose average salary is greater than ₹40,000 using HAVING.
5. Categorize employees using CASE:
    - Salary greater than or equal to 70,000 → High Salary
    - Salary between 50,000 and 69,999 → Medium Salary
    - Salary below 50,000 → Low Salary
6. Display employee names in uppercase and extract the year from their joining date.
7. Display employee name and bonus. If bonus is NULL, display 0 using COALESCE.
8. Find the second-highest salary using a subquery.

Question 2 – Employee Management and Reporting Hierarchy

20 Marks
Organization Management Scenario

Scenario:

The HR department wants to understand the reporting structure of employees and identify managers, their reporting employees, and department-level employee details.

Tasks:

1. Display each employee's name along with their manager's name using a SELF JOIN.
2. Find employees who earn more than their manager's salary.
3. Display all employees, including employees who are not assigned to any department, using an appropriate OUTER JOIN.
4. Find employees who work in the same department as Arun using a nested subquery.
5. Generate an employee hierarchy using a RECURSIVE CTE with the following columns:
    - emp_id
    - emp_name
    - manager_id
    - hierarchy_level
6. Display employees whose salary is greater than ANY salary of employees working in department 3.
7. Find employees who belong to departments where ALL employees earn more than ₹30,000.
8. Display the employee count for each department.

Question 3 – Project and Budget Analysis

20 Marks

Project Management Scenario

Scenario:

A company wants to analyse project allocation, employee participation, project budgets, and project completion status.

The management team wants to identify high-budget projects and compare employee salaries with project budgets.

Tasks:

1. Display employee name, department name, project name, project budget, and project status using joins.
2. Find employees who are not assigned to any project using a LEFT JOIN.
3. Find projects whose budget is greater than the average project budget using a subquery.
4. Display the total project budget for each employee using GROUP BY.
5. Display project details with the following conditions using CASE:
    - Budget greater than or equal to 400000 → High Budget
    - Budget between 200000 and 399999 → Medium Budget
    - Budget below 200000 → Low Budget
6. Find all possible combinations of employees and projects using CROSS JOIN.
7. Display the project name and the number of days since the project started using a date function.
8. Retrieve the second-highest project budget using a nested subquery.

Question 4 – Advanced Employee Data Analysis

20 Marks
Data Analyst Scenario

Scenario:

A business analytics team wants to identify top-performing employees based on salary, department, and joining date.

The company wants to generate a ranking report for management.

Tasks:

1. Rank employees based on salary using the RANK() window function.
2. Display employee name, salary, department ID, and department-wise salary ranking using PARTITION BY.
3. Find the employees who earn more than the overall average salary using a subquery.
4. Display the highest-paid employee in each department using a CTE.
5. Use ROW_NUMBER() to assign a unique ranking to employees based on salary in descending order.

6. Find the difference between each employee's salary and the previous employee's salary using LAG().
7. Display employees who joined after January 1, 2022, and whose salary is between ₹40,000 and ₹70,000.
8. Display employee names starting with 'A' or 'P' using LIKE and combine the results with another city-based condition using logical operators.

Question 5 – Business Database Operations and Management

20 Marks
SQL Developer Scenario

Scenario:

A company wants to perform database maintenance and business reporting. The database administrator must update employee information, handle NULL values, and maintain data consistency.

Tasks:

1. Add a new column called `employment_status` to the employees table with a default value of 'Active'.
2. Update the salary of employees working in the IT department by increasing it by 10%.
3. Display employees whose bonus is NULL and replace the NULL bonus with 0 in the output.
4. Find employees whose salary is not between ₹40,000 and ₹70,000.
5. Display employee details sorted by salary in descending order and limit the output to the top three employees.
6. Find employees who are working in departments located in Chennai or Bangalore using a JOIN and IN operator.
7. Use a CTE to calculate the average salary and display employees earning more than that average.
8. Demonstrate a transaction using START TRANSACTION, UPDATE, ROLLBACK, and COMMIT. Explain the difference between ROLLBACK and COMMIT.
9. Create a VIEW that displays employee name, department name, salary, and location.
10. Explain how PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, DEFAULT, and CHECK constraints help maintain database integrity. Write an example of a CHECK constraint that restricts salary to a positive value.